2 diabetes. "The lifestyle of a stressed person is not healthy for insulin management. They tend to eat less or binge on junk food and have large gaps between meals. They may also falter on exercise and are likely to sleep poorly," notes Dr Mishra.

On other parts of the body, such as the heart, muscles, brain and reproductive system, stress operates in more telling ways. Heart risk is what concerns experts the most. In 2021, a study by researchers from Linköping University in Sweden found that cortisol levels were markedly higher in their respondents in the months before a heart attack; in 2004, INTERHEART, a global case-control study, noted that financial stress increased the risk of myocardial infarction by 13 times in the 106 South African patients who participated. "We are finding the risk to be greater than we previously thought," says Dr Seth. "There are mechanisms that are well-recognised today as results of stress—changes in the inner lining of the heart and increased risk of blood clots. Both can cause sudden heart attack or heart failure." If one has only stress, the risk of heart disease is higher but reversible if caught in time. However, in combination with other risk factors such as hypertension or diabetes, the risk increases nine-fold, according to Dr Seth.

Stress can also induce loss of muscle mass and joint ailments subsequently. Pro-inflammatory cytokines that result from chronic stress can precipitate osteoporosis, for instance. "A dysfunction of the cortisol hormone has a major impact on bone and muscle health," says Dr Sunil Dachepalli, a senior orthopaedic surgeon at Yashoda Hospitals, Hyderabad. "Patients lose muscle mass and put on weight, which puts pressure on the spine and knees in particular.





**"The pandemic has increased stress, which is having a greater impact on health than before. There is an uptick in sudden heart attacks, for example"**

**–DR ASHOK SETH,**
*Chairman,*
*Fortis Escorts Heart Institute*

Those who have reduced bone formation as a result of high inflammation can then have joint pain and problems in mobility; they can even end up with osteoarthritis."

The brain isn't immune to stress either. In 2015, a study involving 138,782 participants published in the journal *Neurology* found that work-related stress in demanding jobs increased the risk of brain stroke, particularly in women. "It increases the risk of blood clots, which can then travel to the brain and cause a stroke," says Dr Renjen. In addition, stress can affect memory. Finally, the dysfunction of one hormone can have a cascading effect on others, affecting fertility and hormonal health as well. "There is a body-mind interlink between neurochemicals and hormones," says Dr Sameer Malhotra, director of the department of mental health and behavioural sciences at the Max hospital in Saket, Delhi. "If you have chronic stress, it can directly disturb dopamine, serotonin and various reproductive hormones, which then causes conditions like depression, low libido, infertility, polycystic ovary syndrome."

## NEW STRESS-BUSTERS

A lot of the recent research is focusing on nuances in how the brain reacts to threats and how this can be used to treat stress. On October 24, Iowa University researchers confirmed for the first time that stress responses were entirely in our brain's control and that manipulating a certain neural pathway could actually improve our coping mechanism. In a study published in the journal *Proceedings of the National Academy of Sciences* (PNAS), they identified the neural circuit in the brain that became active under stress. This finding holds great promise for the treatment and sup-